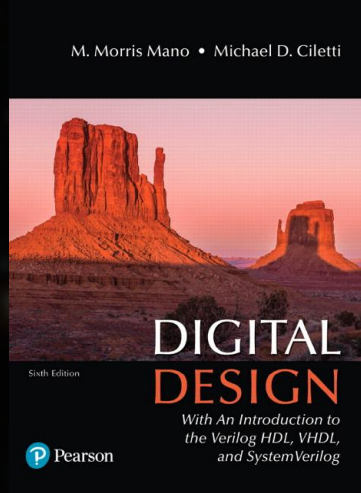
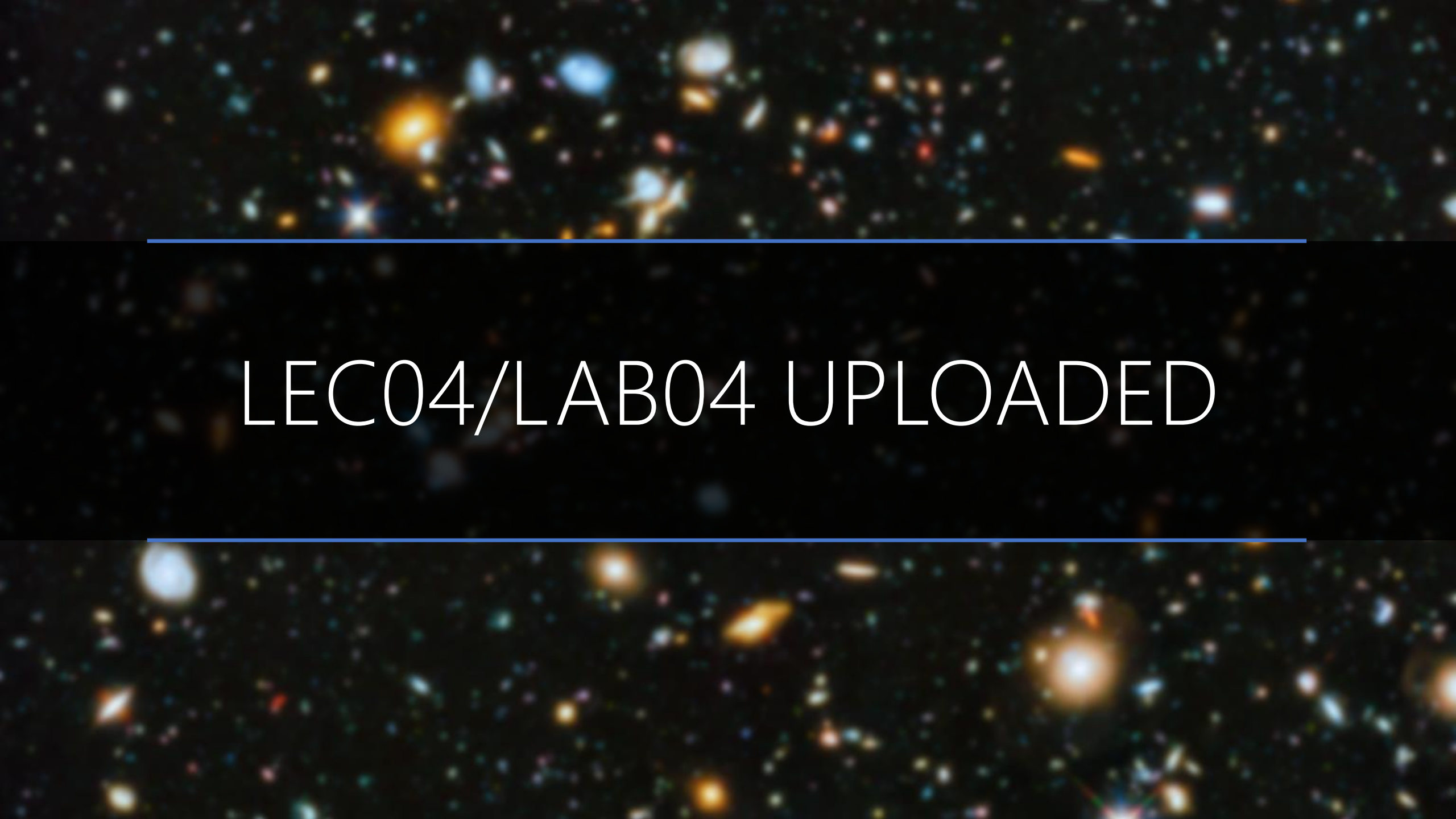




Chapter 1

Digital Systems and Binary Numbers

The background of the slide is a deep space image showing a vast field of galaxies. The galaxies are of various shapes and sizes, some appearing as bright, fuzzy blobs while others are more distinct. The colors range from warm oranges and yellows to cool blues and purples, set against a dark, star-speckled sky. Two thin, horizontal blue lines are positioned above and below the central text, framing it.

LEC04/LAB04 UPLOADED

A deep space image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines are positioned above and below the central text.

CONVERSION

From Base-r to Base-r'

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

ARITHMETIC

A cosmic background image featuring a dense field of galaxies in various colors (yellow, orange, blue, red) against a black space. Two horizontal blue lines are positioned above and below the central text.

ADDITION

PADDING

+ Base-16	0	2	A
	4	B	F

.
.
.

E	5	4
2	B	0

+ Base-16	0	2	A
	4	B	F

.	1		
	E	5	4
	2	B	0
		0	4

$$\frac{16}{16} = 1 \text{ } r \text{ } 0$$

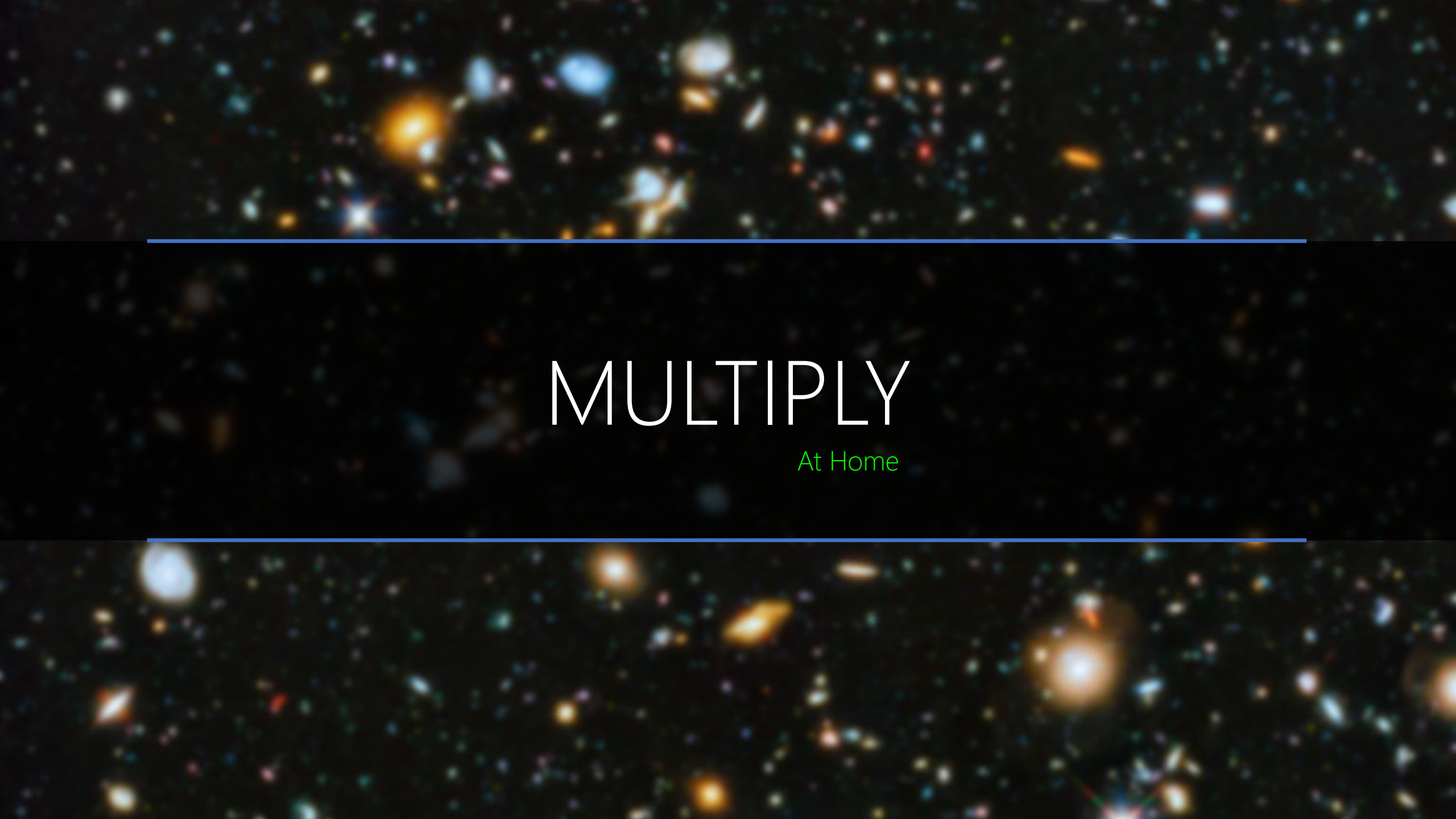
		1	1
+ Base-16	0	2	A
	4	B	F
			A

.

.

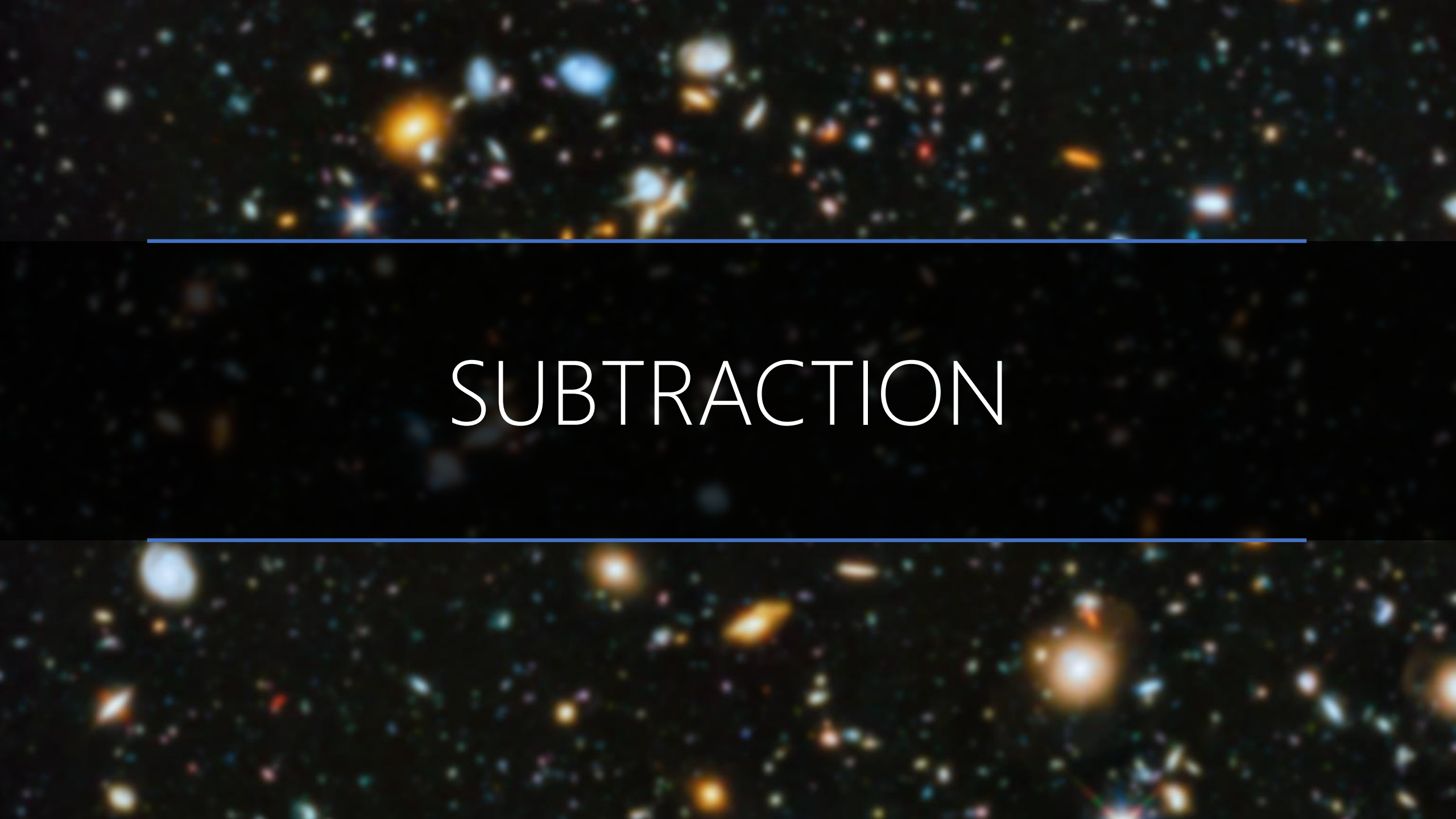
.

1		
E	5	4
2	B	0
1	0	4

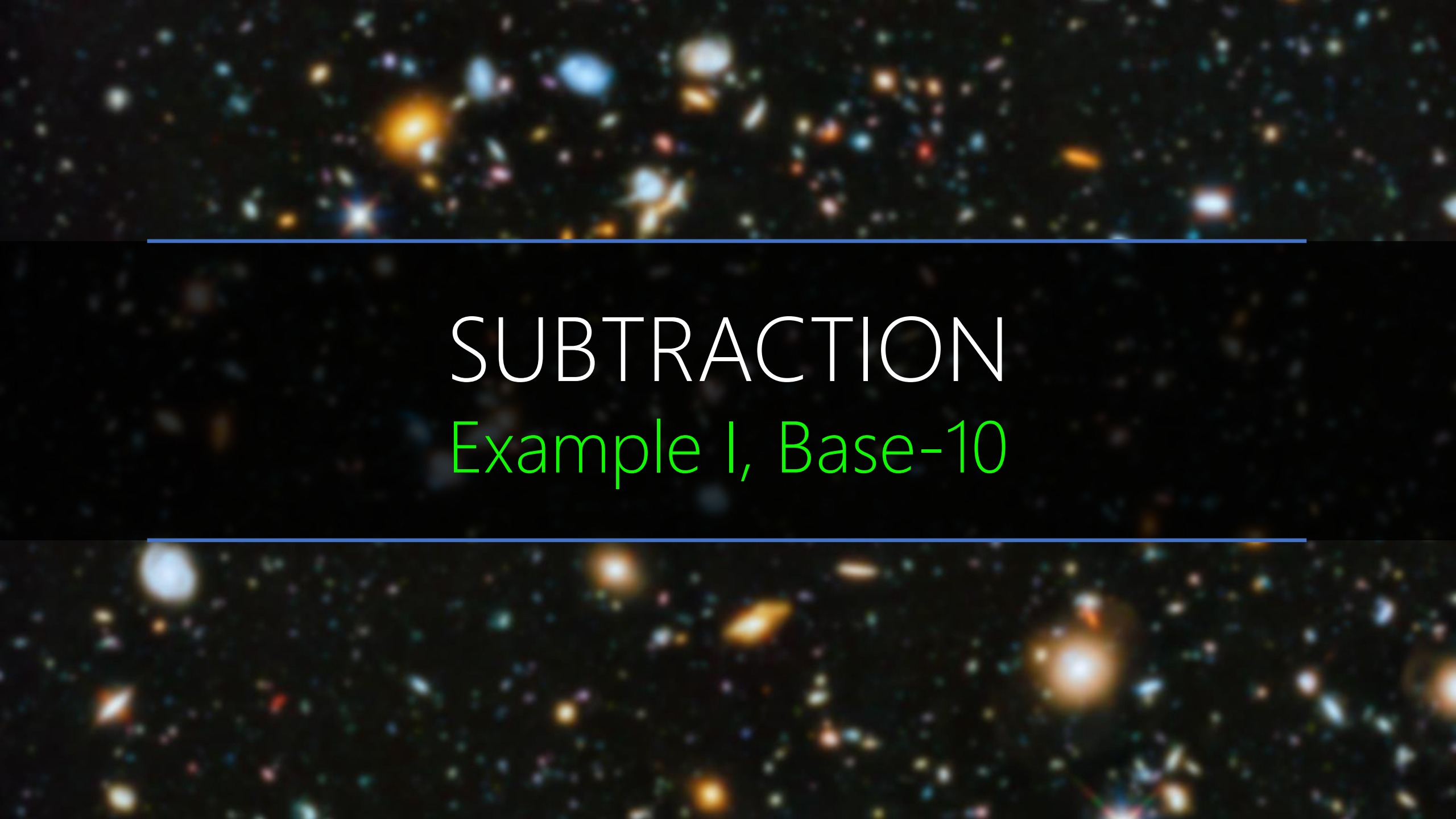
A deep-field astronomical image showing a vast field of galaxies in various colors (yellow, orange, blue, red) against a black background. Two horizontal blue lines are positioned above and below the central text.

MULTIPLY

At Home

A deep space image showing a vast field of galaxies and stars against a black background. The galaxies are in various colors, including blue, orange, and white, and are scattered across the frame. Two horizontal blue lines are positioned above and below the central text.

SUBTRACTION

The background of the slide is a deep space image showing a dense field of galaxies. These galaxies appear as bright, colorful spots (yellow, orange, blue, and white) against a dark, black background. They are scattered across the entire frame, with some appearing larger and more prominent than others. Two thin, horizontal blue lines are positioned above and below the central text, spanning most of the width of the image.

SUBTRACTION

Example I, Base-10

PADDING

— Base-10	0	2	1
	4	3	2

.
.
.

1	5	4
0	6	0

IS IT POSSIBLE?

— Base-10	0	2	1
	4	3	2

.
.
.

1	5	4
0	6	0

				BORROW	
—	0	2	1	.	
Base-10	4	3	2	.	
				.	

-1	→ +10	
1	5	4
0	6	0
	= 10+5-6 = 9	4

		+10	
	-1	-1	+10
—	0	2	1
Base-10	4	3	2
		8	9

.

.

.

-1	+10	
1	5	4
0	6	0
0	9	4

	+10	+10	
-1	-1	-1	+10
— Base-10	0	2	1
	4	3	2
	5	8	9

.

.

.

-1	+10	
1	5	4
0	6	0
0	9	4

	+10	+10	
-1	-1	-1	+10
—	0	2	1
Base-10	4	3	2
	5	8	9

.
.
.

-1	+10	
1	5	4
0	6	0
0	9	4

Last Borrow → 021.154 < 432.060 → Negative Result

	+10	+10	
-1	-1	-1	+10
— Base-10	0	2	1
	4	3	2
	5	8	9

.

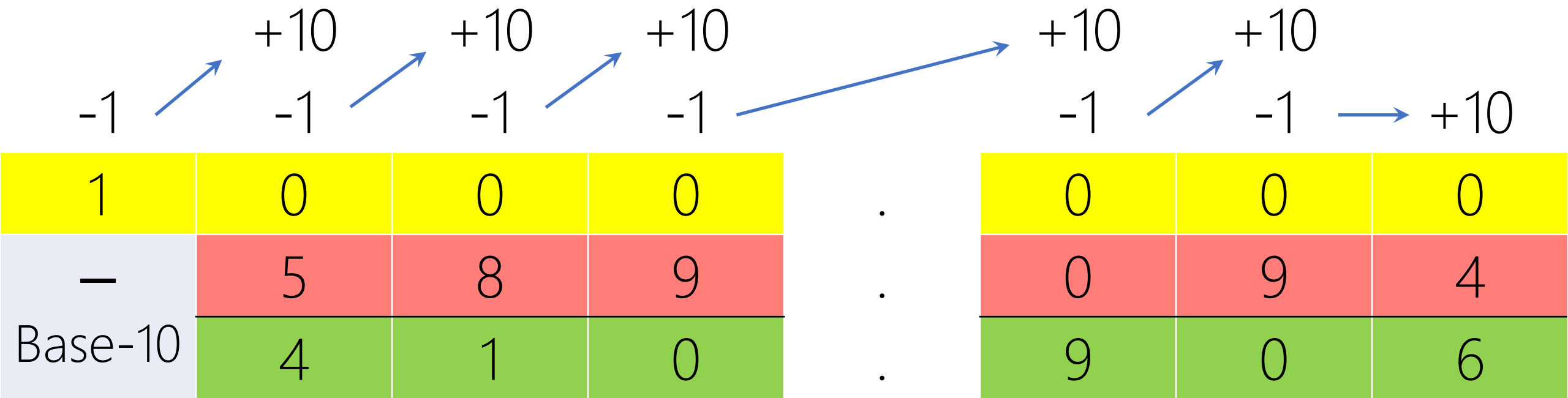
.

.

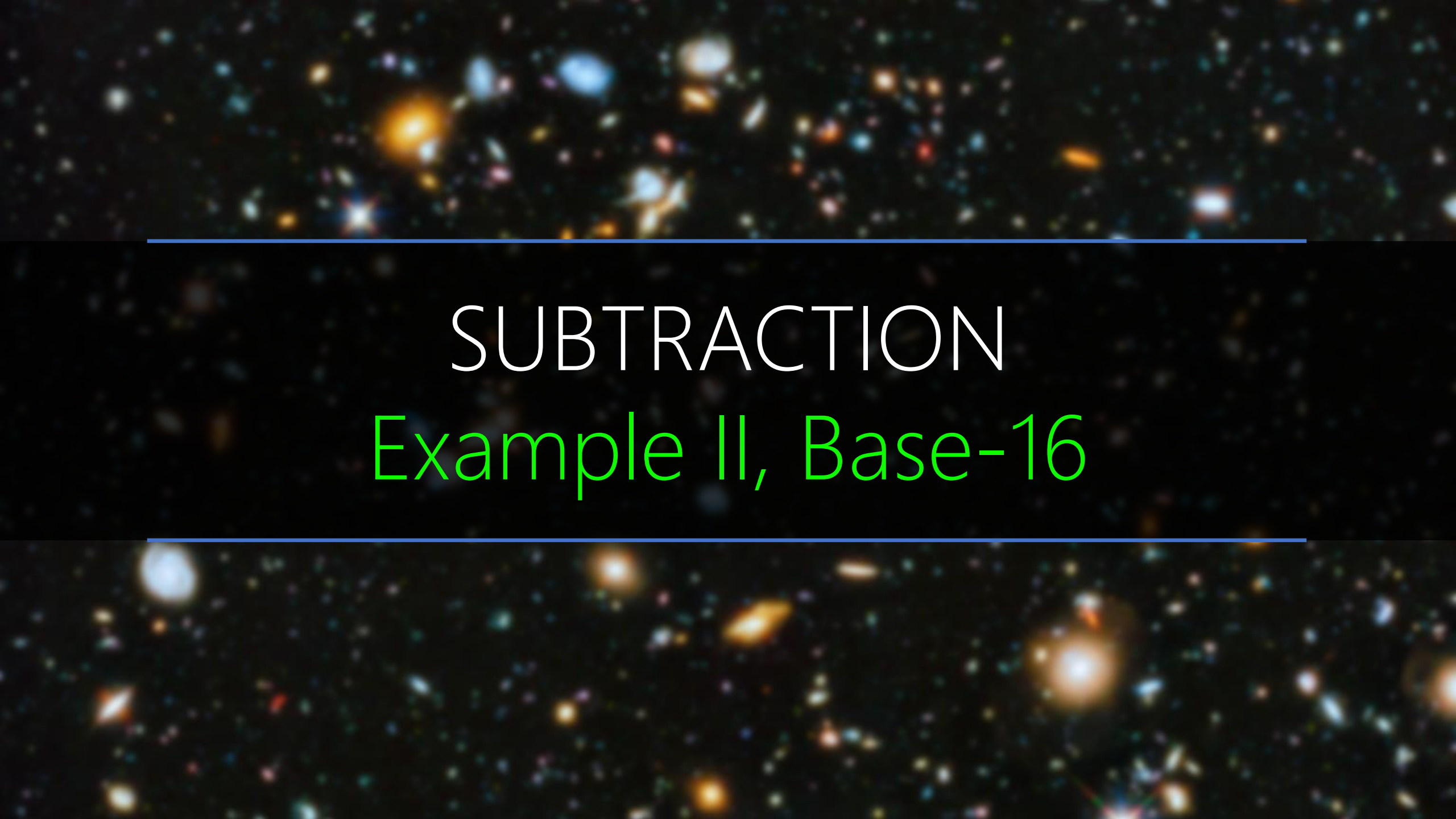
-1	+10	
1	5	4
0	6	0
0	9	4

The Result is not correct.

Extra Subtraction:



$$\begin{aligned} &= (021.154)_{10} - (432.060)_{10} = (021.154)_{10} + (1000.000)_{10} - (1000.000)_{10} - (432.060)_{10} \\ &= - (1000.000)_{10} + (589.094)_{10} = - [(1000.000)_{10} - (589.094)_{10}] \\ &= - (410.906)_{10} \end{aligned}$$

The background of the slide is a deep space image showing a dense field of galaxies. These galaxies appear in various colors, including blue, orange, and white, against a black background. Two horizontal blue lines are positioned above and below the text.

SUBTRACTION

Example II, Base-16

PADDING

— Base-16	0	2	A
	4	B	F

.
.
.

E	5	4
2	B	0

				BORROW	
—	0	2	A	.	
Base-16	4	B	F	.	
				.	

-1	→ +16	
E	5	4
2	B=11	0
	= 5+16-11 = 10 = A	4

— Base-16	0	2	A
	4	B	F

.	-1	+16	
.	E=14	5	4
.	2	B	0
.	B=11	A	4

		+16	
	-1	-1	+16
—	0	2	A
Base-16	4	B=11	F
		6	B

.

.

.

-1	+16	
E	5	4
2	B	0
B	A	4

		+16	
	-1	-1	+16
—	0	2	A
Base-16	4	B	F
		6	B

.

.

.

-1	+16	
E	5	4
2	B	0
B	A	4

	+16	+16	
-1	-1	-1	+16
— Base-16	0	2	A
	4	B	F
	B=11	6	B

.

.

.

-1	+16	
E	5	4
2	B	0
B	A	4

	+16	+16	
-1	-1	-1	+16
—	0	2	A
Base-16	4	B	F
	B	6	B

.

.

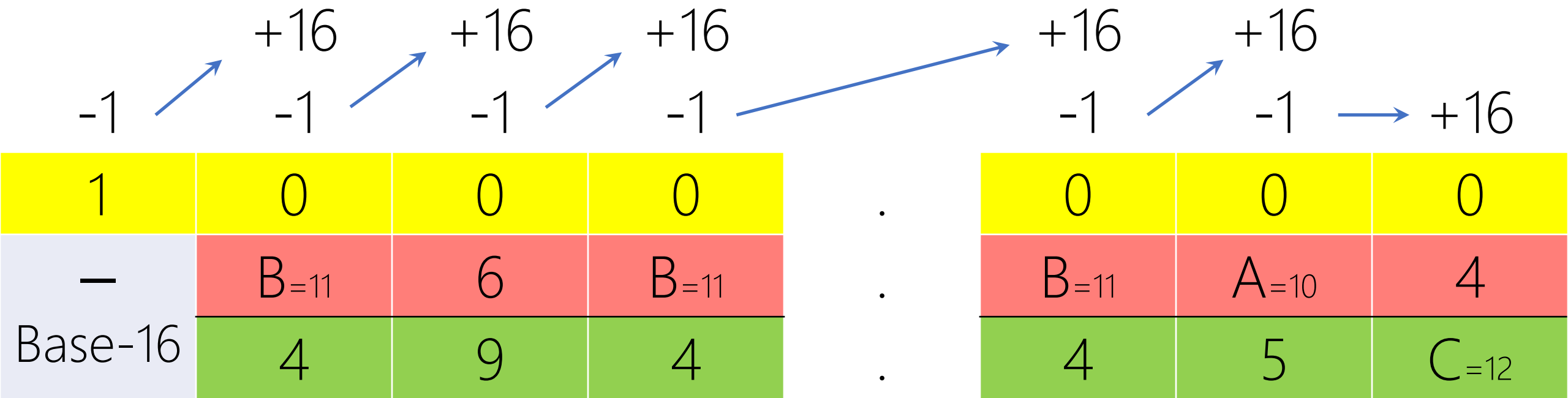
.

-1	+16	
E	5	4
2	B	0
B	A	4

02A.E54 < 4BF.2B0

Last Borrow → Negative Result

Extra Subtraction:



Calculation steps:

$$\begin{aligned} &= (2A.E54)_{16} - (4BF.2B0)_{16} = (2A.E54)_{16} + (1000.000)_{16} - (1000.000)_{16} - (4BF.2B0)_{16} \\ &= -(1000.000)_{16} + (B6B.BA4)_{16} = -[(1000.000)_{16} - (B6B.BA4)_{16}] \\ &= -(494.45C)_{16} \end{aligned}$$

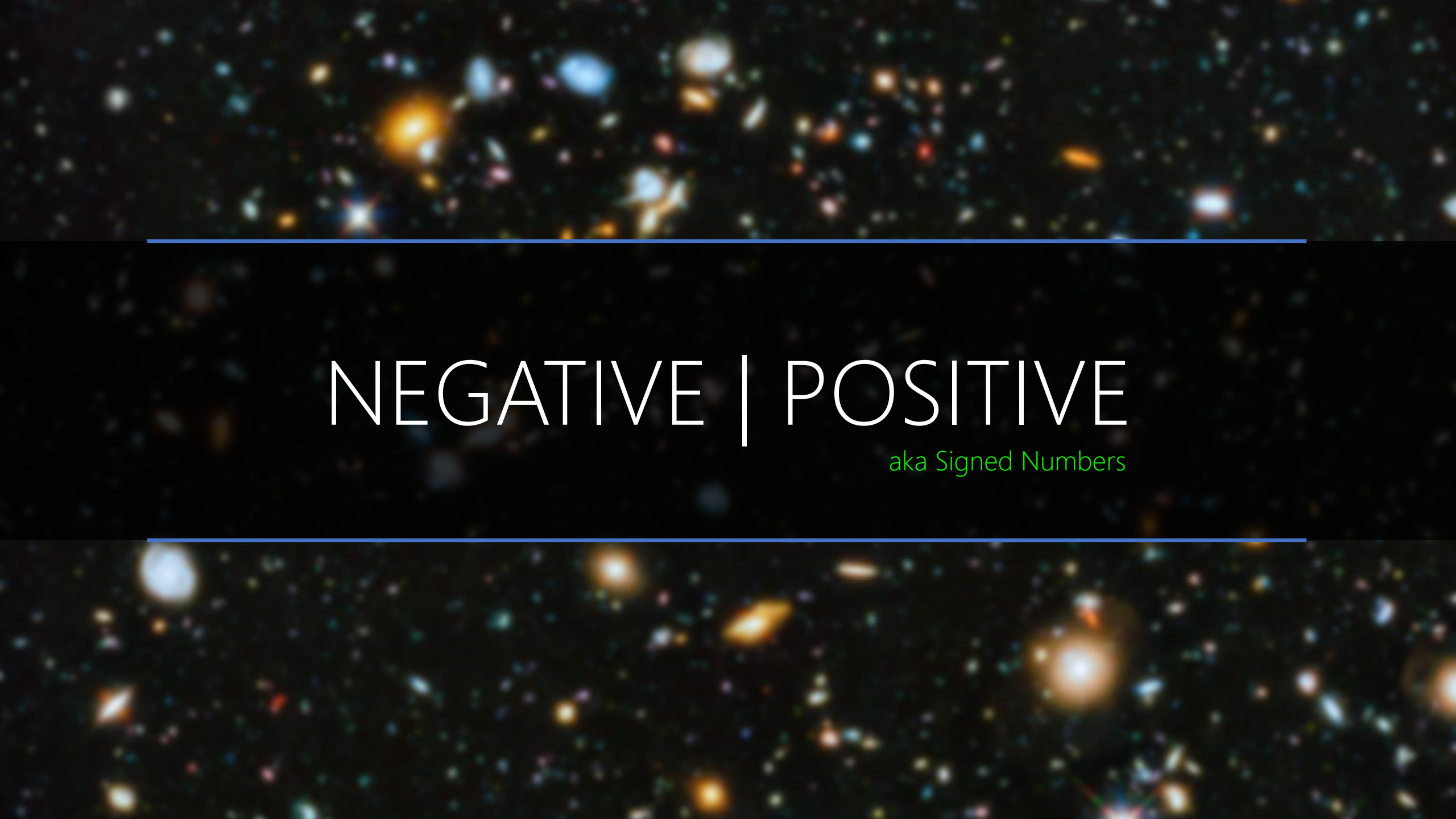
The background of the entire image is a deep space photograph showing a vast field of galaxies and stars. The galaxies are in various stages of evolution, some appearing as bright, smooth ellipses and others as more complex, irregular shapes. The stars are small, distant points of light scattered across the dark void. The overall color palette is dominated by the blues and oranges of the galaxies, with the white and yellow of the stars providing contrast.

DIVISION

At Home

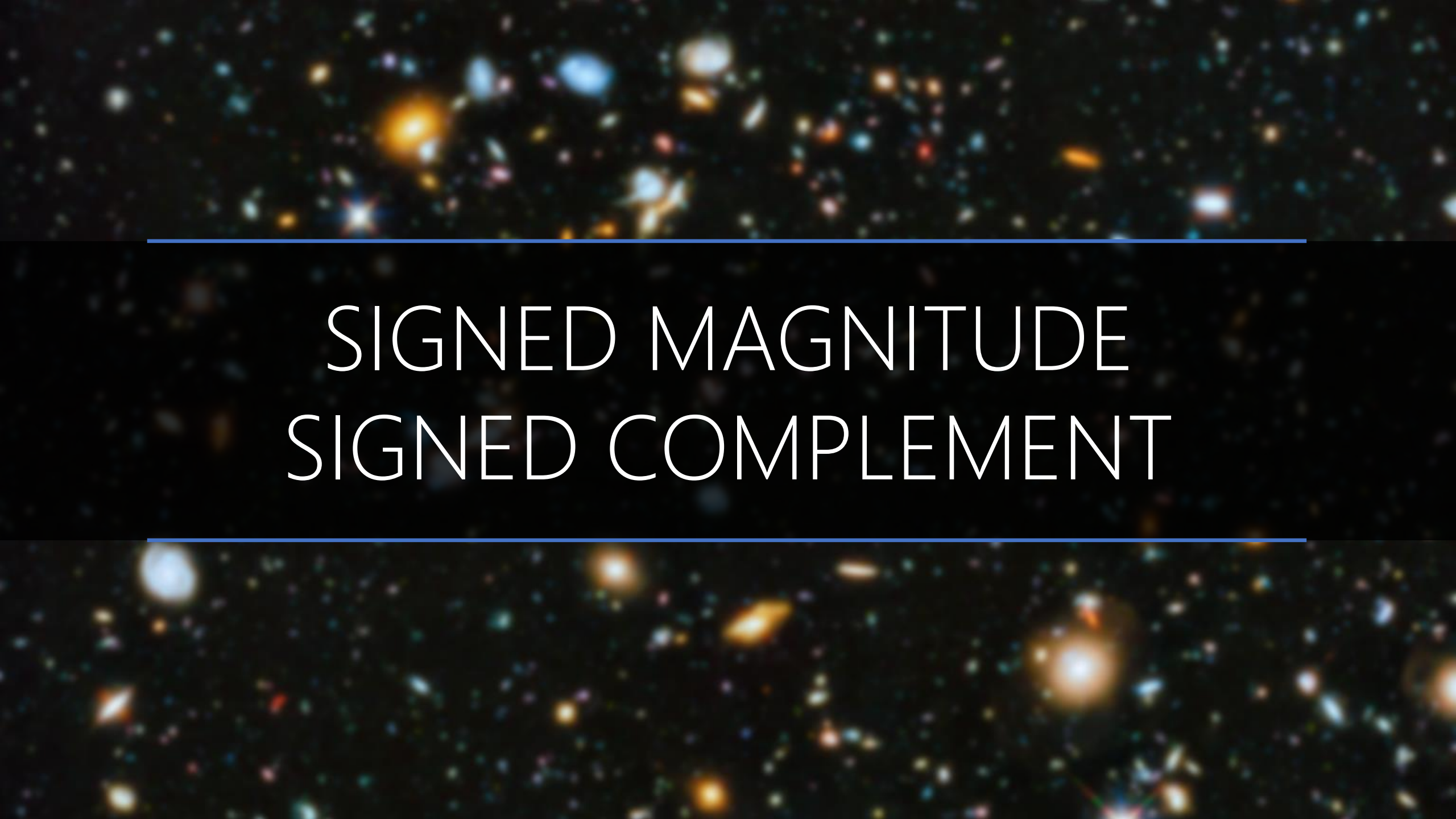
A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

SIGNED NUMBERS

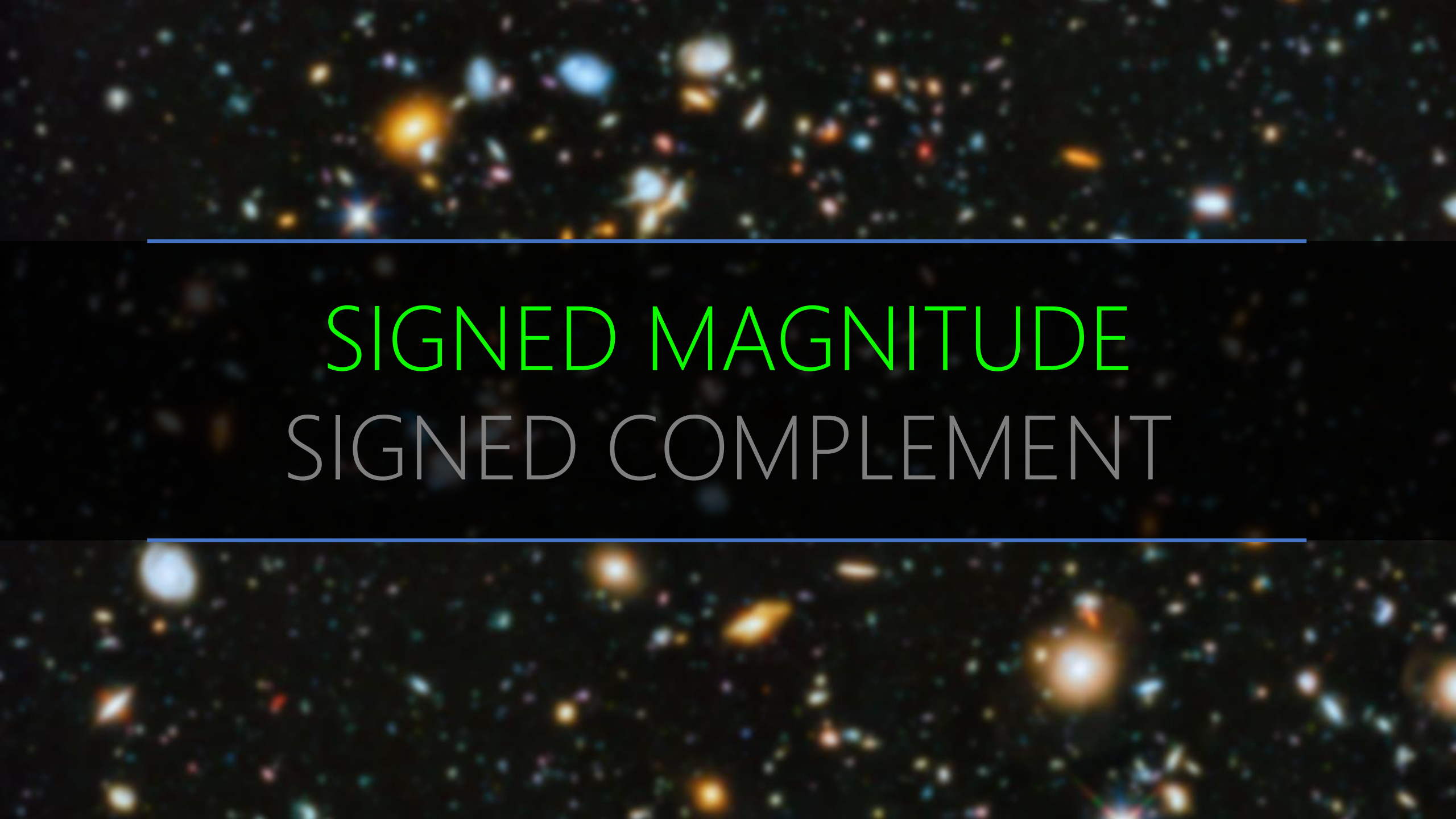
A cosmic background image featuring a dense field of galaxies in various colors (blue, orange, white) against a black space. Two horizontal blue lines are positioned above and below the central text.

NEGATIVE | POSITIVE

aka Signed Numbers

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, diffuse clouds and others as more compact, point-like sources. The image is used as a background for a title slide.

SIGNED MAGNITUDE
SIGNED COMPLEMENT

A deep space image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

SIGNED MAGNITUDE
SIGNED COMPLEMENT

10^0

0

1

2

3

4

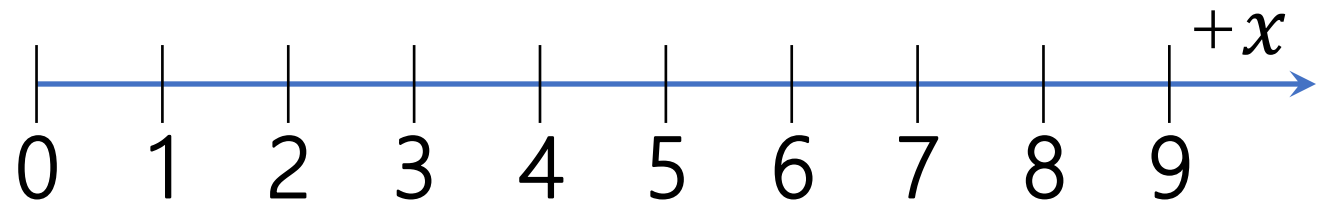
5

6

7

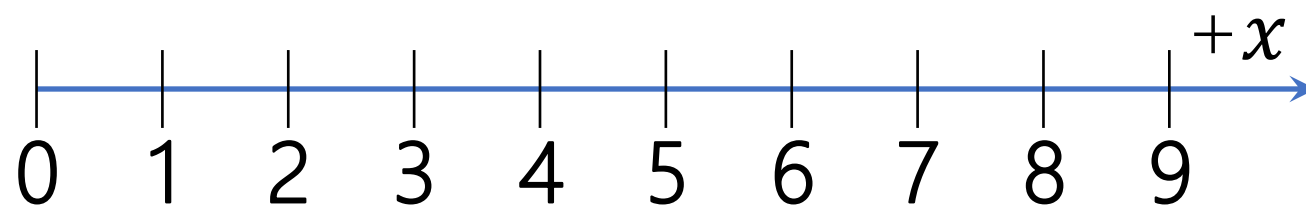
8

9



Any idea to represent negative numbers?

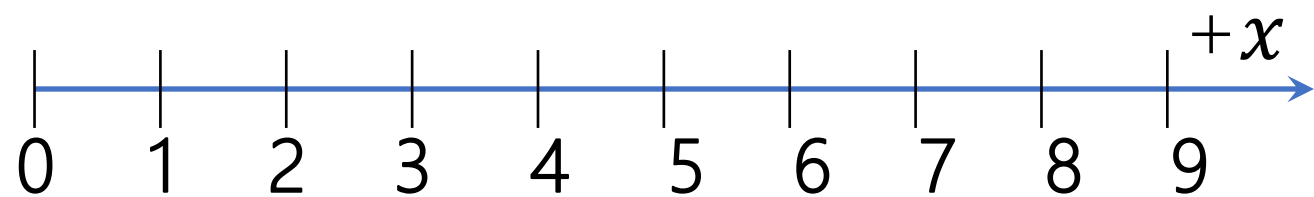
10^0
0
1
2
3
4
5
6
7
8
9



New symbols for each digit?

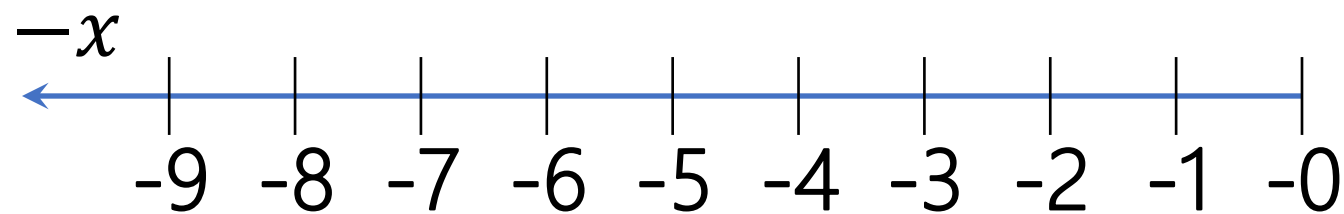
↑ ∑ ε ∴ ∂ ∇ 8 9

10^1	10^0
	0
	1
	2
	3
	4
	5
	6
	7
	8
	9



A new position?

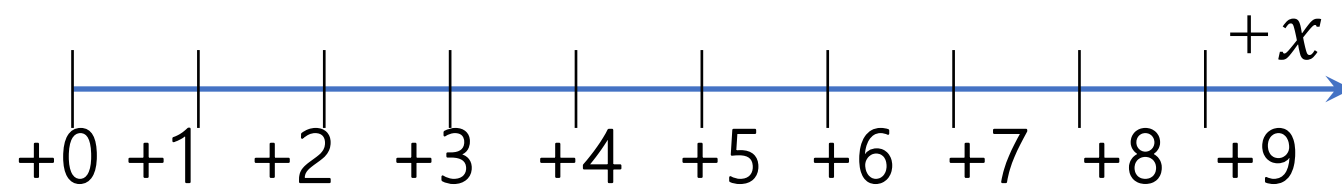
10^1	10^0
—	0
—	1
—	2
—	3
—	4
—	5
—	6
—	7
—	8
—	9



A new position? Yes

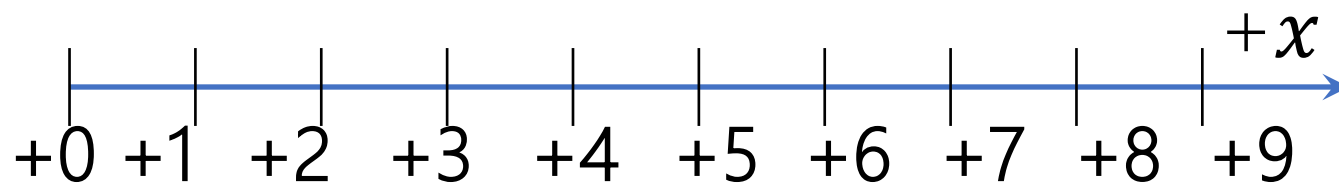
A new symbol: —

10^1	10^0
+	0
+	1
+	2
+	3
+	4
+	5
+	6
+	7
+	8
+	9



A new position? Yes
A new symbol: +

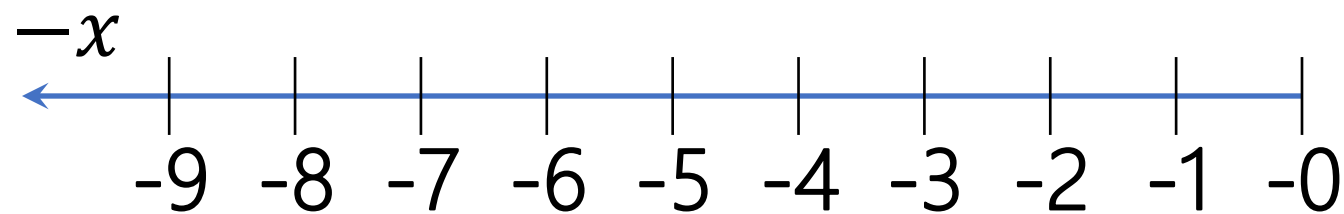
10^1	10^0
0	0
0	1
0	2
0	3
0	4
0	5
0	6
0	7
0	8
0	9



A new position? Yes

A new symbol: $+$

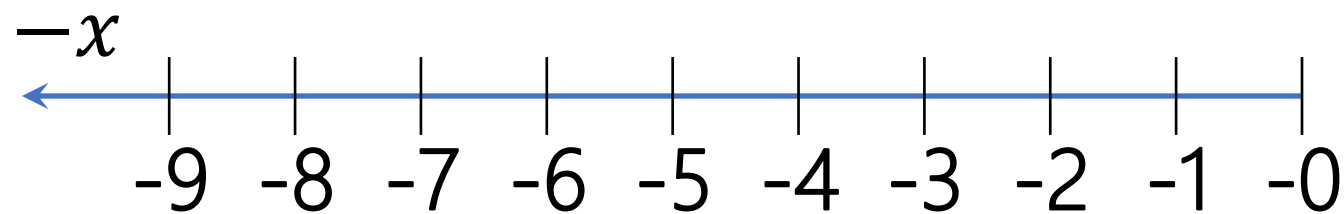
10^1	10^0
1	0
1	1
1	2
1	3
1	4
1	5
1	6
1	7
1	8
1	9



A new position? Yes

A new symbol: $-$

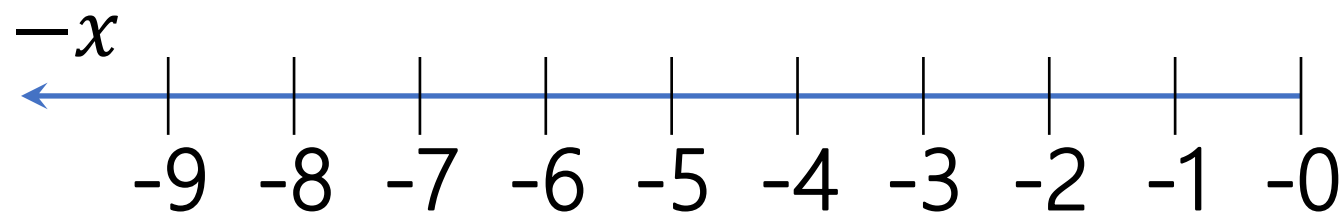
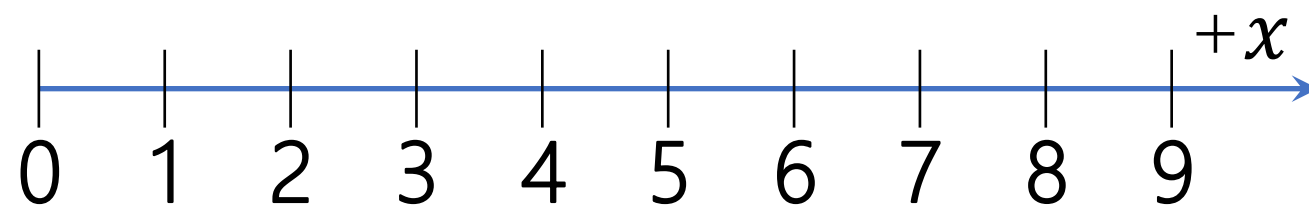
10^1	10^0
Nonzero	0
Nonzero	1
Nonzero	2
Nonzero	3
Nonzero	4
Nonzero	5
Nonzero	6
Nonzero	7
Nonzero	8
Nonzero	9



A new position? Yes

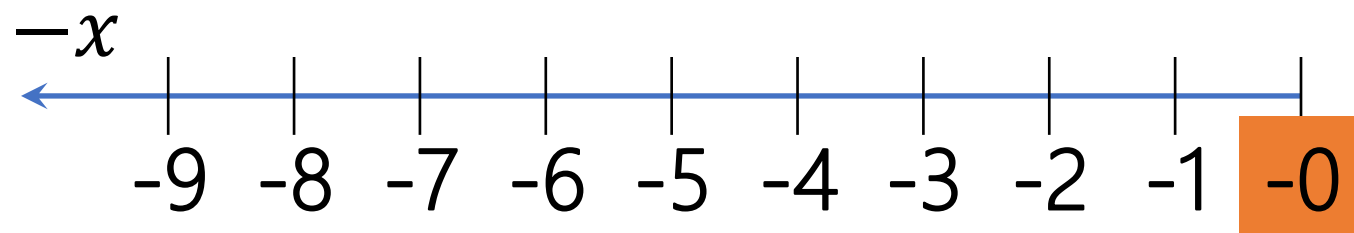
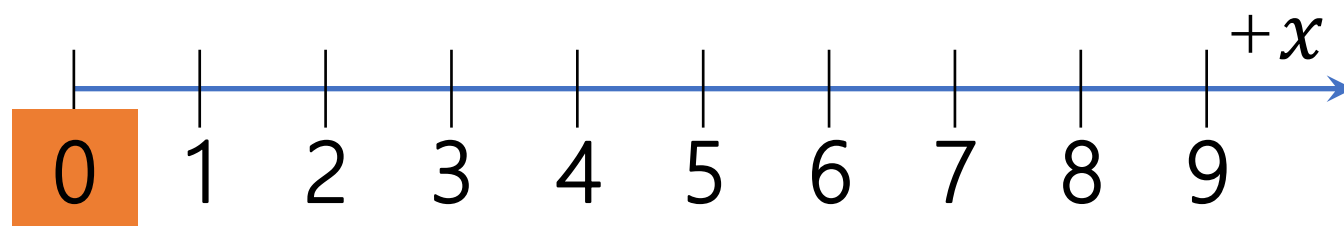
A new symbol: $-$

10^1	10^0
1,2,...9	0
1,2,...9	1
1,2,...9	2
1,2,...9	3
1,2,...9	4
1,2,...9	5
1,2,...9	6
1,2,...9	7
1,2,...9	8
1,2,...9	9

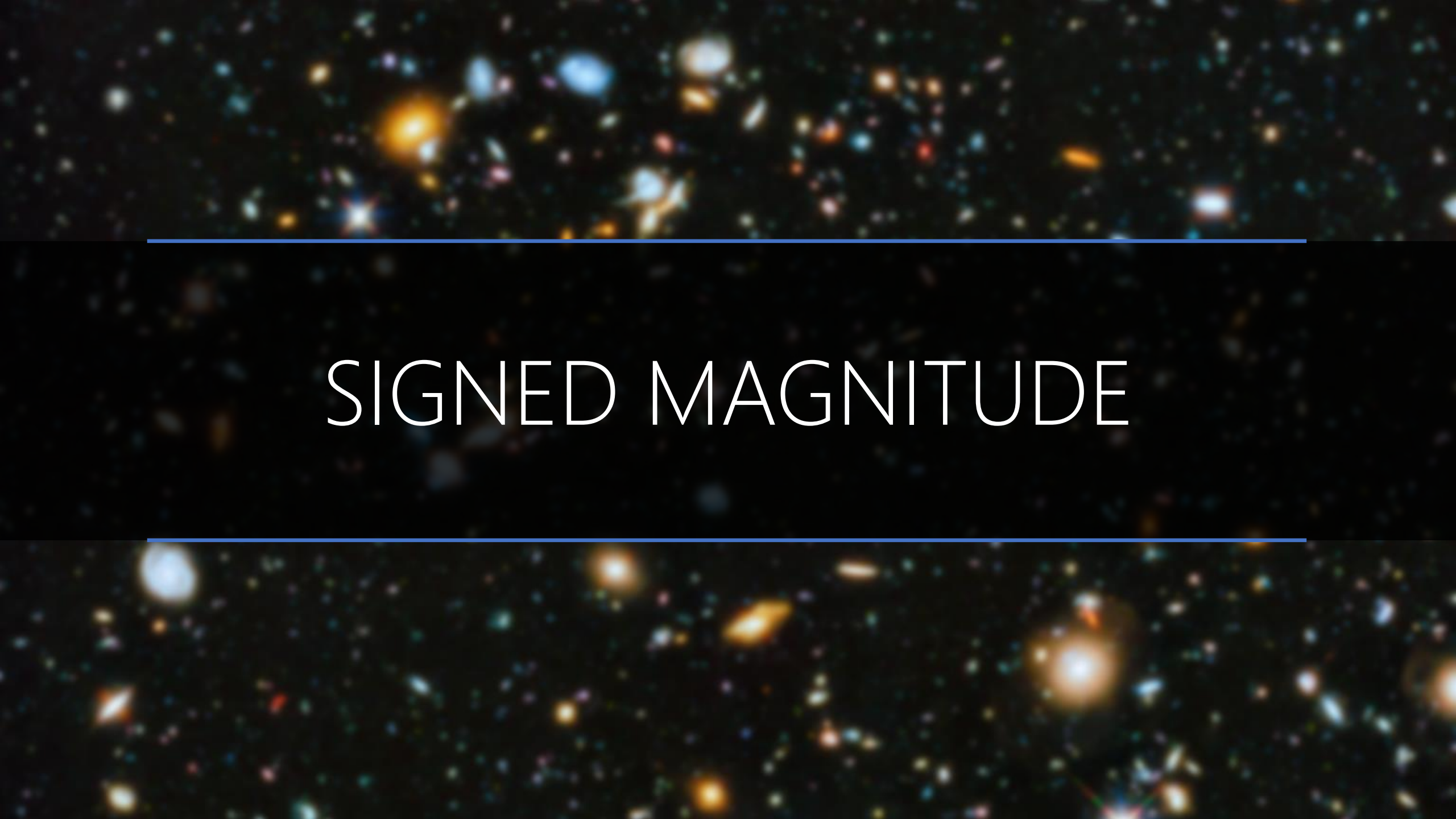


10^1	10^0
0	0
0	1
0	2
0	3
0	4
0	5
0	6
0	7
0	8
0	9

10^1	10^0
1,2,...9	0
1,2,...9	1
1,2,...9	2
1,2,...9	3
1,2,...9	4
1,2,...9	5
1,2,...9	6
1,2,...9	7
1,2,...9	8
1,2,...9	9



10^1	10^0
0	0
0	1
0	2
0	3
0	4
0	5
0	6
0	7
0	8
0	9

The image features a deep space background filled with numerous galaxies of various shapes and colors, including yellow, orange, and blue, set against a black void. Two thin, horizontal blue lines are positioned above and below the central text. The text "SIGNED MAGNITUDE" is centered in a white, serif, all-caps font.

SIGNED MAGNITUDE

r^{n-1}	r^{n-2}	r^{n-3}	...	r^2	r^1	r^0
0	Positive Numbers					
Nonzero	Negative Numbers					



Signed

Magnitude

Give up left most position for sign!

r^{n-1}	r^{n-2}	r^{n-3}	...	r^2	r^1	r^0
0	Positive Numbers					
Nonzero	Negative Numbers					

$$\text{Min} = -(r^{n-1} - 1) \leftarrow \cancel{0} \rightarrow \text{Max} = r^{n-1} - 1 = \cancel{r^n - 1}$$

2^5	2^4	2^3	2^2	2^1	2^0
0	0	0	0	1	1
1	0	0	1	0	1

Interpretation	Base-10	
?	?	?
?	?	?

7^5	7^4	7^3	7^2	7^1	7^0
0	0	0	0	5	6
1	0	0	2	0	5
4	0	0	2	0	5
6	0	0	2	0	5

Interpretation	Base-10			
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?

2^5	2^4	2^3	2^2	2^1	2^0
0	0	0	0	1	1
1	0	0	1	0	1

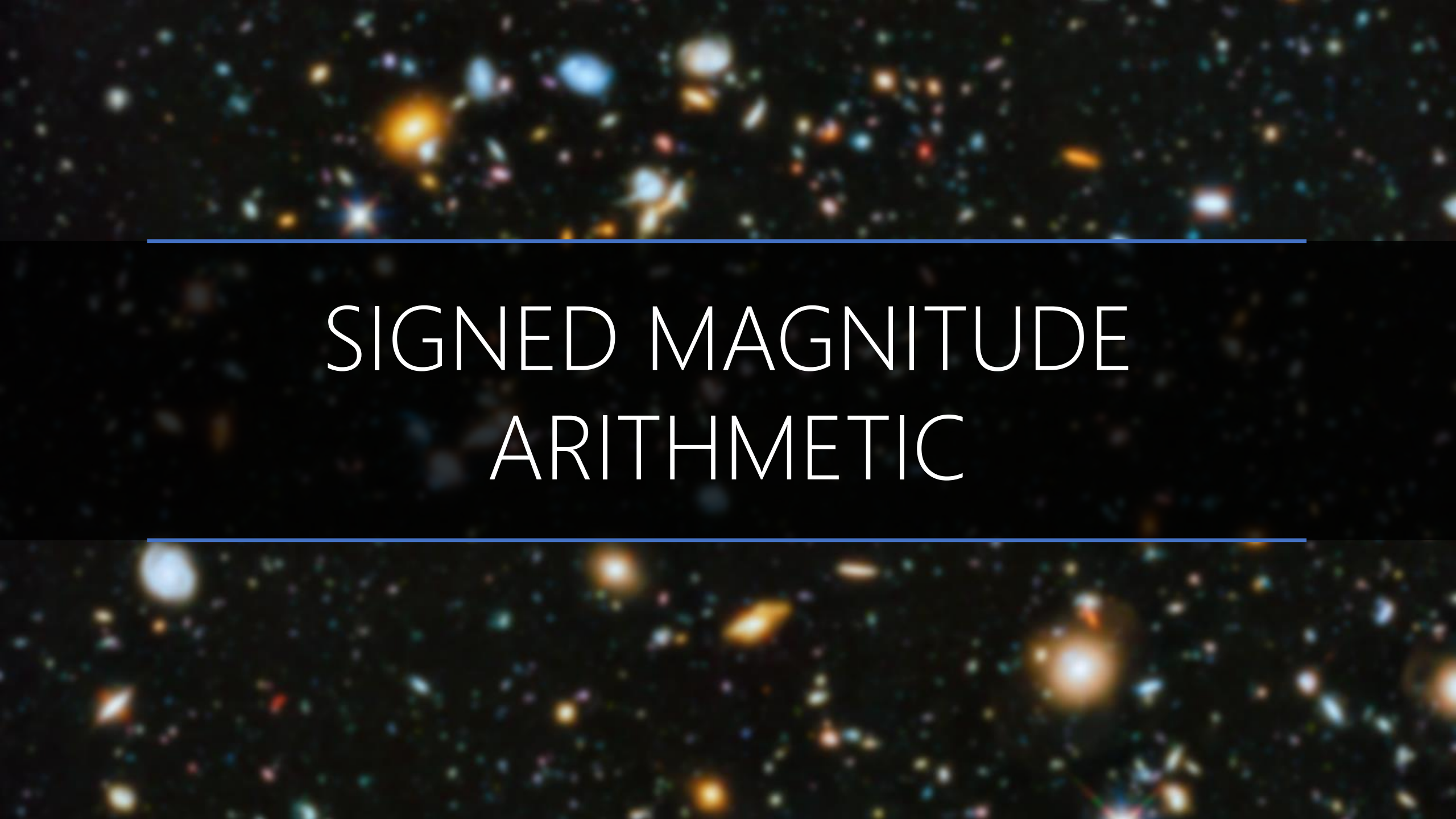
Interpretation	Base-10	
$+(3)_{10}$	0	3
$-(5)_{10}$	1,2,...9	5

7^5	7^4	7^3	7^2	7^1	7^0
0	0	0	0	5	6
1	0	0	2	0	5
4	0	0	2	0	5
6	0	0	2	0	5

Interpretation	Base-10			
$+(41)_{10}$	0	0	4	1
$-(103)_{10}$	1,2,...9	1	0	3
$-(103)_{10}$	1,2,...9	1	0	3
$-(103)_{10}$	1,2,...9	1	0	3



There Is No Symbol For $-$, $+$
Just Our Interpretation!

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

SIGNED MAGNITUDE ARITHMETIC

0	X
0	X
1	X

+

0	Y
1	Y
1	Y

=

0	$X+Y$
$X < Y$ (if borrow)	$X-Y$ (if borrow, apply it)
1	$-(X+Y) = X+Y$

+

=

+

=

0	X
0	X
1	X

-

0	Y
1	Y
1	Y

=

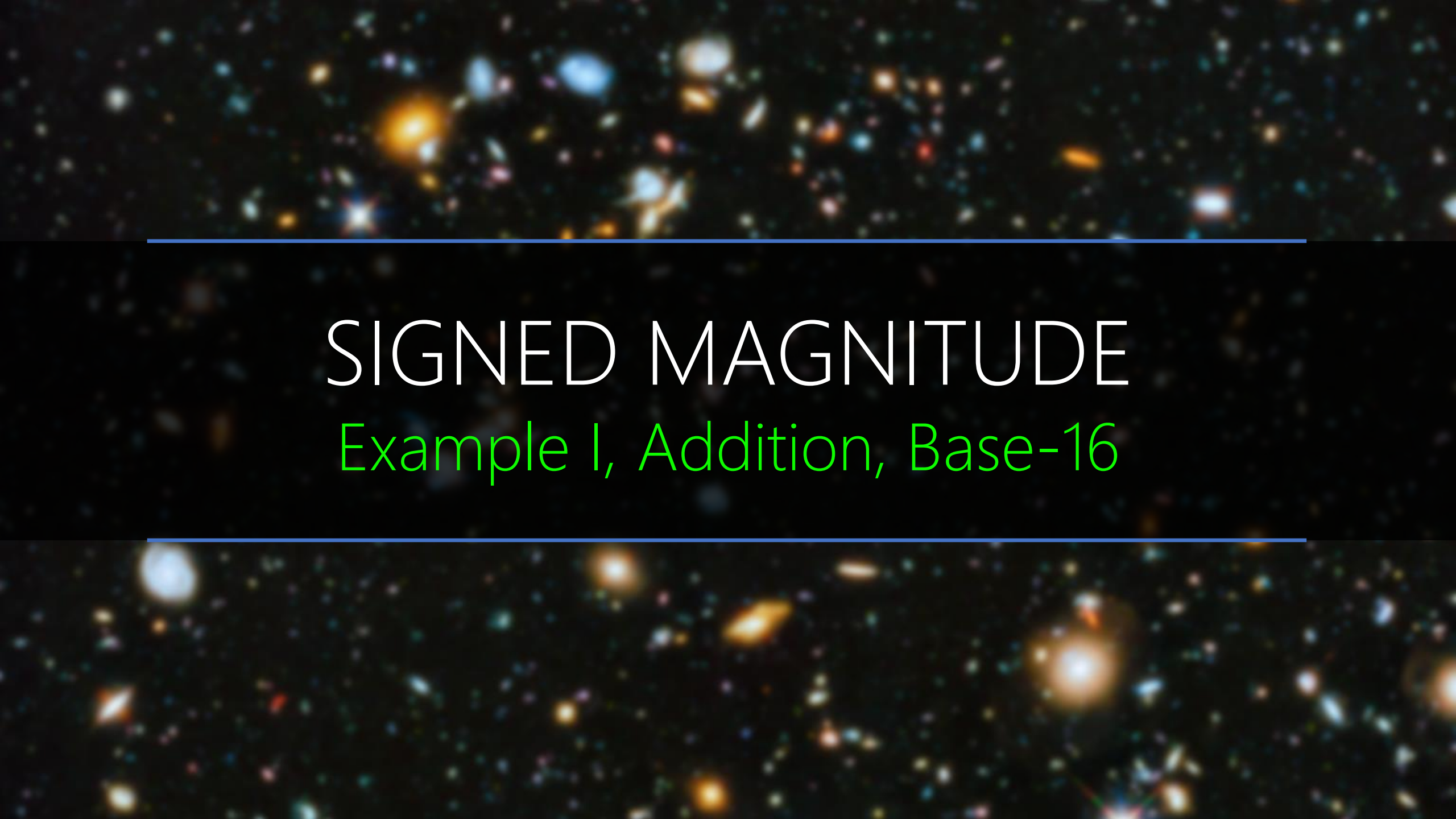
$X < Y$ (if borrow)	$X-Y$ (if borrow, apply it)
0	$X+Y$
$X > Y$ (if borrow)	$-X+Y = Y-X$ (if borrow, apply it)

-

=

-

=

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

SIGNED MAGNITUDE

Example 1, Addition, Base-16

+ Base-16	1	2	A	.	E	5	4
	1	B	F	.	2	B	
				.			

+ Base-16	1	2	A	.	E	5	4
	1	B	F	.	2	B	
				.			

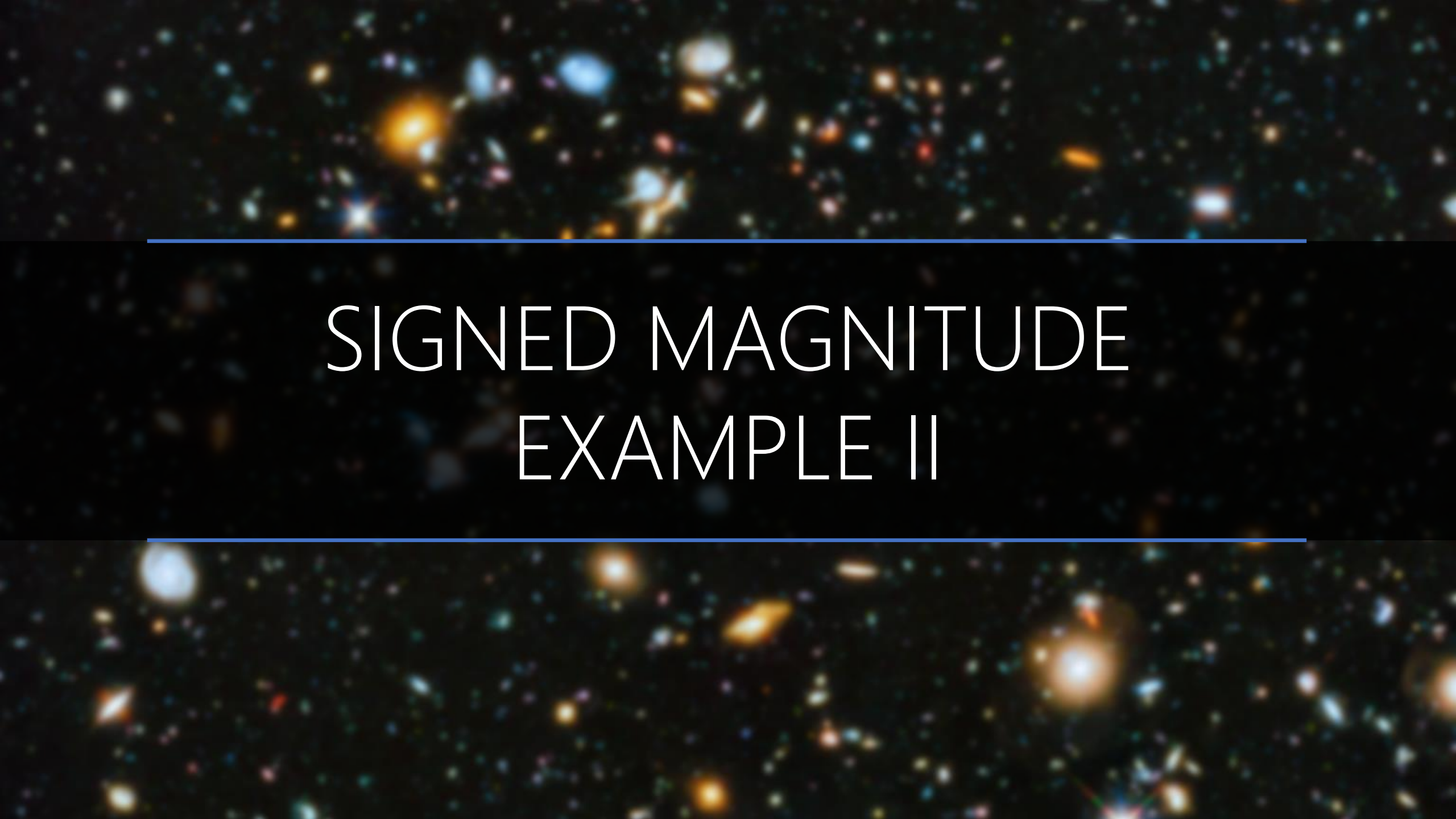
PADDING

+ Base-16	1	2	A	.	E	5	4
	1	B	F	.	2	B	0
				.			

SIGNED: $(-X) + (-Y) = -(X + Y)$

+ Base-16	1	2	A	.	E	5	4
	1	B	F	.	2	B	0
	1			.			

+ Base-16		1	1		1		
	1	2	A ₌₁₀	.	E ₌₁₄	5	4
	1	B	F ₌₁₅	.	2	B ₌₁₁	0
	1	E ₌₁₄	A ₌₁₀	.	1	0	4

The background of the slide is a deep space image showing a vast field of galaxies. These galaxies appear as bright, colorful spots and streaks of light against a dark, black background. The colors range from bright yellow and orange to deep blues and purples, representing different wavelengths of light. The galaxies are scattered across the entire frame, with some appearing larger and more prominent than others. Two thin, horizontal blue lines are positioned above and below the central text, framing it.

SIGNED MAGNITUDE EXAMPLE II

+ Base-16	0	2	A	.	E	5	4
	1	B	F	.	2	B	
				.			

+ Base-16	0	2	A	.	E	5	4
	1	B	F	.	2	B	
				.			

PADDING

+ Base-16	0	2	A	.	E	5	4
	1	B	F	.	2	B	0
				.			

SIGNED: $(+X) + (-Y) = ?(X - Y)$

Base-16	0	2	A	.	E	5	4
	1	B	F	.	2	B	0
	?			.			

—
Base-16

Last Borrow $\rightarrow$ Negative Result

		+16	+16		+16	+16	
	-1	-1	-1		-1	-1	+16
— Base-16	1	0	0	.	0	0	0
		6	B=11	.	B=11	A=10	4
		9	4	.	4	5	C=12

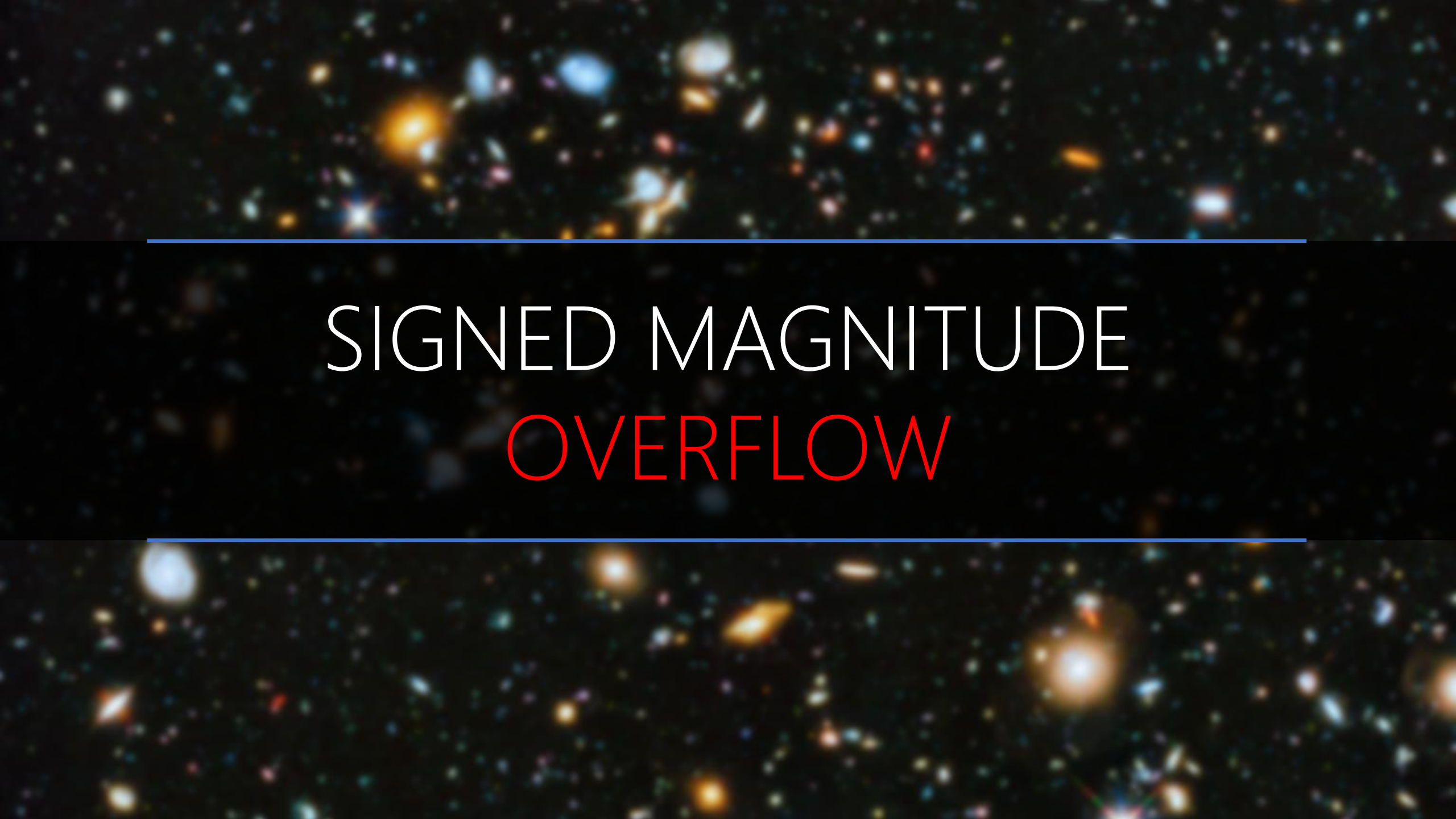
$$\begin{aligned}
 &= (2A.E54)_{16} - (BF.2B0)_{16} = (2A.E54)_{16} + (100.000)_{16} - (100.000)_{16} - (BF.2B0)_{16} \\
 &= -(100.000)_{16} + (6B.BA4)_{16} = -[(100.000)_{16} - (6B.BA4)_{16}] \\
 &= -(94.45C)_{16}
 \end{aligned}$$

		-1	+16					
			-1	+16			-1	+16
Base-16	0	2	A ₌₁₀	.		E ₌₁₄	5	4
	1	B ₌₁₁	F ₌₁₅	.		2	B ₌₁₁	0
	1	6	B ₌₁₁	.		B ₌₁₁	A ₌₁₀	4

2A.E54 < BF.2B0

Last Borrow → Negative Result

1	9	4	.	4	5	C
---	---	---	---	---	---	---

The background of the slide is a deep space image showing a vast field of galaxies. These galaxies appear as bright, colorful spots and streaks of light against a dark, black background. The colors range from bright yellow and orange to deep blues and purples, representing different wavelengths of light. The galaxies are scattered across the entire frame, with some appearing larger and more prominent than others. Two thin, horizontal blue lines are positioned above and below the central text, framing it.

SIGNED MAGNITUDE OVERFLOW

Overflow!

0	X
0	X
1	X

+

0	Y
1	Y
1	Y

=

0 → 1	X+Y
X<Y	X-Y
1 → 1	X+Y

0	X
0	X
1	X

-

0	Y
1	Y
1	Y

=

X<Y	X-Y
0 → 1	X+Y
X>Y	-X+Y=Y-X

— Base-2	$+(27)_{10}$	0	1	1	0	1	1
		1	0	1	1	1	0

— Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$-(14)_{10}$	1	0	1	1	1	0

$+X - (-Y)$

— Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$-(14)_{10}$	1	0	1	1	1	0

$+X - (-Y) = +(X+Y)$

Base-2	+(27) ₁₀	0	1	1	0	1	1
	+(14) ₁₀	0	0	1	1	1	0

$+X - (-Y) = + (X + Y)$

Base-2	+ $+(27)_{10}$	0	1	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
		0					

$+X - (-Y) = +(X+Y)$

Base-2 +		1	1	1	1		
	+(27) ₁₀	0	1	1	0	1	1
	+(14) ₁₀	0	0	1	1	1	0
		0	0	1	0	0	1

$$+X - (-Y) = +(X+Y)$$

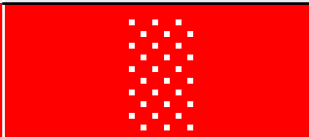
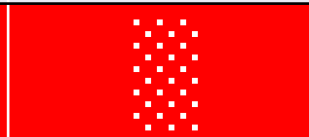
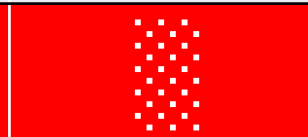
Base-2		1	1	1	1		
	+(27) ₁₀	0	1	1	0	1	1
	+(14) ₁₀	0	0	1	1	1	0
		1	0	1	0	0	1

If you consider the last carry $\rightarrow -(9)_{10} \rightarrow$ Negative!

$$+X - (-Y) = +(X+Y)$$

		1	1	1	1		
+ Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
		0	0	1	0	0	1
		If you ignore it $\rightarrow +(9)_{10} \rightarrow$ Result is not correct!					

$$+X - (-Y) = +(X+Y)$$

		1	1	1	1		
+ Base-2	$+(27)_{10}$	0	1	1	0	1	1
	$+(14)_{10}$	0	0	1	1	1	0
							
		Overflow: The result is not reliable!					

PRACTICE

1. Base (r)
2. First Number (X)
3. Second Number (Y)
4. Signed or Unsigned?
 1. If Signed, What Signed System?
5. Operation?
6. Overflow?

PRACTICE (chatgpt?)

1. Base ($r=7$)
2. First Number ($X=023.31$)
3. Second Number ($Y=121.46$)
4. Signed or Unsigned? Yes
 1. If Signed, What Signed System? Signed Magnitude
5. Operation? Subtraction
6. Overflow?

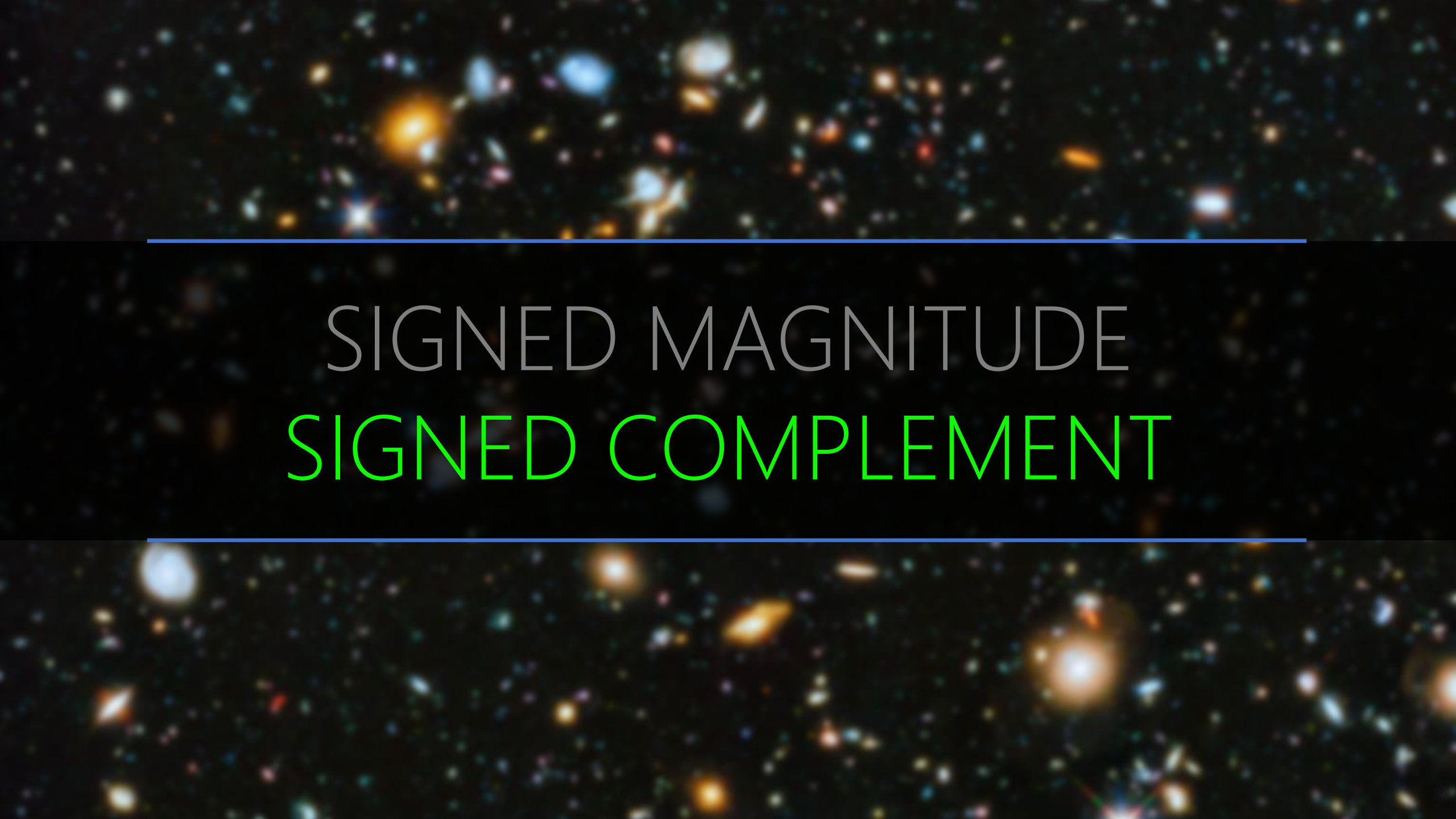
A cosmic background image featuring a dense field of galaxies in various colors (yellow, orange, blue, and red) against a dark space. Two horizontal blue lines are positioned above and below the central text.

WHY NOT SIGNED MAGNITUDE

Give up left most position for sign! What are the wastes?

r^{n-1}	r^{n-2}	r^{n-3}	...	r^2	r^1	r^0
0	Positive Numbers					
Nonzero	Negative Numbers					

$$\begin{aligned} &+0 \rightarrow \text{Max} = r^{n-1} - 1 = \cancel{r^n - 1} \\ \text{Min} &= -(r^{n-1} - 1) \leftarrow -0 \end{aligned}$$

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

SIGNED MAGNITUDE
SIGNED COMPLEMENT